Emily Fang

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EDUCATION

North Carolina State University, Raleigh, NC
Ph.D. in Industrial Engineering (Human Factors & HCI), GPA 3.94 — *Aug 2024 – Present*
B.S. in Industrial Engineering, Minor in Supply Chain Management, GPA 3.77 — *Aug 2019 – May 2024*

RESEARCH EXPERIENCE

North Carolina State University, Graduate Research Assistant — *May 2024 – Present*

Short-Term Cognitive Effects of Different Video Game Tasks | *Lead Researcher* — *May 2024 – Present*

- Independently designed and ran a three-condition lab experiment (n = 46, 2-hour sessions) testing whether 30 minutes of FPS, strategy, or narrative gameplay changes attention, working memory, and multitasking.
- Uncovered a gap between how players felt and how they performed: FPS players rated time pressure 2.4× and effort 1.9× higher than the control group, and every FPS player said the game affected them, yet measured performance did not differ.
- Combined 20,000+ task trials, eye tracking, NASA-TLX and PANAS surveys, and interviews; analyzed with ANCOVA in R and thematic analysis of all 46 interviews. First-authored an HFES 2026 paper.

AI-Assisted AR Checklists for Construction Hazard Recognition | *Lead Researcher & Developer* — *Aug 2025 – Oct 2026*

- Designed and built a hands-free Meta Quest 3 app (Unity/C#) that sends the headset's camera view to GPT-4o and places an OSHA hazard-mitigation checklist in view, applying human factors principles to panel placement, system status, and "Verify" steps.
- Benchmarked 10 prompt versions against 42 expert-annotated images, raising precision from 63% to 75% and cutting false alarms by 44% without losing recall. Hosted a 3-hour hands-on demo at HFES 2026; code on GitHub.

Scale Cognition Advanced Learning Environments in VR (SCALE-VR) | *Qualitative Researcher* — *Aug 2023 – Jul 2025*

- Shaped three design changes to an NSF-funded VR learning app (replacing grabbing with automatic stacking, removing the "under view," and recoloring a hard-to-see ramp) through thematic analysis of 30 think-aloud sessions and 15 interviews.
- Presented findings weekly to an interdisciplinary team led by 3 PIs (engineering, design, education), and coded A/B preference tests with 10 usability experts to help weigh usability against learning-theory goals.
- First-authored a think-aloud study (n = 15) showing students relied most on the numeric height panel while none mentioned using color or sound cues; presented a 20-minute talk at HFES 2025.

North Carolina State University, Undergraduate Research Assistant — *Aug 2021 – May 2024*

Augmented Reality for Engineering Education Enhancement (AREEA) | *Research Assistant* — *Oct 2022 – May 2024*

- Helped run all phases of a semester-long study (n = 42) of a HoloLens 2 game-based learning app, logging task times and misclicks, administering PSSUQ and bipolar laddering interviews, grading assessments, and coding interviews; the study-design paper won 1st Place, Best New Career Paper Award at ASEE 2025.

Virtual Instructor Application 2 (VIA 2) | *Qualitative Researcher* — *Aug 2021 – Oct 2022*

- Coded interview and bipolar laddering data from 13 workers testing an AR posture-training app as one of two coders, flagging easy-to-miss cues and unresponsive gestures; the redesign beat the original on all four PSSUQ usability scales.

SKILLS & HONORS

Research methods: interviews, think-aloud usability testing, A/B preference testing, bipolar laddering, surveys (PSSUQ, NASA-TLX, PANAS), thematic analysis, experimental design, eye tracking
Statistics & tools: ANOVA/ANCOVA, regression, nonparametric tests; R, Python, Qualtrics, NVivo, Dedoose, Figma, Tableau; Unity (C#), Meta Quest 2/3, HoloLens 2, OpenAI API
Honors: NIOSH Graduate Fellowship (2025 – Present), Provost's Doctoral Fellowship (2024 – 2025), IISE Applied Ergonomics Student of the Year (2023)

SELECTED PUBLICATIONS

Cheng, F., **Fang, E.**, & Chen, K. B. (2026). Elicitation and assessment of hand-based gestures for information processing in immersive environments. *Human Factors*.

Fang, E., & Chen, K. B. (2026). Examining short-term video game play on subsequent working memory performance. *Proc. HFES*.

Perera, G. N., **Fang, E.**, Bottomley, L., Chen, K. B., & Ivy, J. (2025). Study design and assessment framework for testing augmented reality tools in engineering education. *ASEE*. **Best New Career Paper, 1st Place.**

Fang, E., Kulasingam, R., Cheng, F., Peterson, M., Delgado, C., & Chen, K. B. (2025). Examining user interactions with signaling elements in a virtual reality learning application. *Proc. HFES*.

Fang, E., Sivaramakrishnan, A., & Chen, K. B. (2024). Negative emotions from virtual reality usage: A preliminary exploratory study using online forums. *Proc. HFES*.

Full list of 8 publications in my CV: ehfang.github.io/CV_EmilyFang.pdf